

SALIENT FAUSTA MAKIA NKONGHO ENOH

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PROFESSIONAL SUMMARY

Robotics & Mechatronics Engineering student with proven experience building ROS2 autonomy stacks, real-time vision pipelines (PyTorch/OpenCV), and sensor fusion systems for ground and underwater robots. Deployed perception on NVIDIA Jetson GPUs, built reproducible data pipelines from sensor logs, and integrated vision modules into multi-sensor autonomy stacks, directly matching Nextpower's Robotics Autonomy Software Internship. Available for Group 1 (May 18) or Group 2 (June 15) start.

EDUCATION

B.S. Robotics, Mechatronics & Automation Engineering | Minor: Computer Science | GPA: 3.65/4.0

Expected May 2027

Widener University | Chester, PA

Honors: Widener Presidential Scholarship, PECO Engineering Scholarship, Dean's List, Honors in Engineering

Relevant Coursework: Robotics, Autonomous Navigation, Computer Vision, Sensor Fusion, MATLAB, C++, Python, Linear Algebra

SKILLS

Languages & Systems: Python, C++, Embedded C, Linux (Ubuntu), Git, Multithreading, Systems Programming

Robotics: ROS2, Gazebo, RViz, SLAM, Path Planning, MATLAB/Simulink, Autonomous Driving Simulation, Scenario Generation, SIL/HIL

Vision & ML: PyTorch, OpenCV, Object Detection, Image Processing, Real-Time Inference, GPU Acceleration, Computer Vision Pipelines

Data & Systems: Data Pipelines, Log Analysis, Real-Time Data Streaming, Deterministic Systems Design, Profiling & Debugging

Hardware: NVIDIA Jetson (GPU), Raspberry Pi, STM32, Arduino, CPU/GPU Integration

Certifications: Universal Robot e-Series Core Track (January 2026)

EXPERIENCE

Widener University Robotics, AI & Automation Research Lab | Chester, PA

Robotics Autonomy Software Researcher - Ground Vehicle Platform

May – Sep 2025

- Designed and deployed a ROS2 autonomy stack (C++/Python) for real-time perception, planning, and control on a semi-autonomous electric wheelchair ground robot
- Built a real-time ROS2 vision node integrating camera, LiDAR, IMU, and ultrasonic data to output obstacle proximity and risk signals
- Implemented sensor fusion (Kalman filtering) for robust localization and trajectory planning under dynamic outdoor-style conditions
- Developed reproducible data pipelines from sensor logs for system validation and performance analysis
- Created simulation environments and scenario generation pipelines (ROS2, MATLAB) for SIL and HIL testing of the autonomy stack
- Deployed perception and control stack on NVIDIA Jetson with GPU acceleration for real-time inference
- Visualized and processed LiDAR point cloud data in RViz for environment modeling and perception validation (SLAM)
- Reduced system inaccuracy and improved reliability by ~75% through iterative profiling, debugging, and optimization
- Awarded Best Research Presentation for delivering a functional autonomous mobility system

Embedded Software & Vision Engineer - Autonomous Underwater Vehicle (RoboSub 2025)

May – Sep 2025

- Developed ROS2-based embedded software (C++/Python, Linux) for real-time navigation and mission execution
- Built vision-based perception pipelines (OpenCV) for underwater object detection under low-visibility conditions
- Applied image enhancement and filtering techniques to improve perception robustness in noisy environments
- Integrated multi-modal sensor data (camera, IMU, pressure) for state estimation and environmental awareness
- Implemented sensor fusion algorithms for robust perception and localization in dynamic underwater conditions
- Performed simulation-based validation (MATLAB/Simulink) and real-world testing to identify and resolve system issues
- Experimented with PyTorch and OpenCV-based vision models for real-time obstacle detection; evaluated model performance against environmental constraints
- Collaborated across multidisciplinary teams in an international robotics competition 2025

Autonomous Firefighting Robot

Jan – May 2025

- Designed and implemented embedded control software (Arduino, C/C++) for autonomous navigation and actuation
- Developed closed-loop motor control (PID + encoder feedback) for precise motion control
- Integrated multi-sensor inputs (thermal + ultrasonic) for real-time environmental interaction and deterministic behavior
- Fabricated mechanical housing components using AutoCAD, FEA analysis, and rapid prototyping

AI & Machine Learning Researcher

May – Sep 2024

- Designed and deployed a multimodal AI Chatbot system integrating computer vision and speech recognition
- Developed vision-based facial recognition and tracking pipelines using camera input to recognize specific users
- Implemented speech-to-text and text-to-speech systems for real-time human interaction
- Integrated machine learning models and APIs, engineered in Python on Linux, optimizing real-time performance and usability

Robotics, Automation & Controls Research Lab Assistant

Jan – May 2024

- Assisted with mechanical assembly and programming of a humanoid robot for human interaction and live demonstrations
- Supported sensor installation, actuator integration, and firmware debugging during robotics system testing

LEADERSHIP & SERVICE

Founder & President, Widener AI Club: Founded and lead a 40+ member organization focused on robotics, AI, and engineering

